

BRITT, ON, Canada

WMO: 727320

Lat: 45.800N

Lon: 80.530W

Elev: 623

StdP: 14.37

Time Zone: -5.00 (NAE)

Period: 86-00

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -23.1	(c) -16.0	(d) -32.6	(e) 0.9	(f) -22.2	(g) -25.9	(h) 1.3	(i) -15.0	(j) 17.6	(k) 21.5	(l) 16.1	(m) 23.1	(n) 1.1	(o) 10	(p) 0.528

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.3	82.7	69.2	80.0	67.6	77.4	66.1	72.1	78.7	70.4	76.5	68.7	74.5	5.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.0	112.9	75.6	68.4	106.6	73.9	66.8	100.8	71.8	36.3	79.0	34.7	76.5	33.3	74.3	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 14.5	(b) 12.7	(c) 11.3	(e) -34.9	(f) 87.7	(g) 6.5	(h) 2.9	(i) -39.6	(j) 89.8	(k) -43.4	(l) 91.4	(m) -47.1	(n) 93.0	(o) -51.8	(p) 95.1		
			DB													
			WB													
				-31.7	75.6	7.5	1.7	-37.1	76.8	-41.5	77.8	-45.7	78.8	-51.1	80.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.6	13.0	13.9	24.9	39.0	52.3	60.5	65.6	63.4	55.0	44.9	31.9	20.3
	DBStd	20.87	14.87	12.88	12.26	7.62	7.42	6.55	5.23	6.06	7.58	7.47	8.72	13.44
	HDD50	5039	1147	1011	778	338	63	4	0	1	38	195	544	920
	HDD65	9096	1612	1431	1243	779	396	165	58	103	306	625	993	1385
	CDD50	1588	0	0	0	9	135	318	483	417	189	36	1	0
	CDD65	170	0	0	0	0	3	29	76	55	7	0	0	0
	CDH74	1475	0	0	0	8	63	289	628	426	60	1	0	0
	CDH80	233	0	0	0	1	3	46	108	67	8	0	0	0
(14) Wind	WSAvg	4.1	5.0	4.5	4.5	4.6	4.0	3.6	3.1	2.9	3.3	4.4	4.7	4.3
(15) Precipitation	PrecAvg	39.0	2.9	2.1	2.2	3.2	3.5	3.3	3.3	3.7	3.9	4.3	3.5	3.0
	PrecMax	46.1	4.7	4.4	5.5	5.8	6.1	5.6	5.4	6.6	7.6	7.0	5.4	6.1
	PrecMin	31.9	1.5	0.8	0.3	1.2	1.3	1.1	1.5	1.9	1.5	1.9	1.6	1.3
	PrecStd	3.8	0.8	0.9	1.1	1.1	1.2	1.1	1.1	1.1	1.4	1.4	1.0	0.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.7	46.0	57.7	70.2	79.2	84.3	87.0	85.5	79.6	69.4	56.4	46.7
		MCWB	45.4	42.0	49.1	57.1	62.0	70.5	69.9	73.0	67.0	59.9	53.4	44.9
	2%	DB	38.4	40.7	50.6	61.9	75.3	81.0	82.8	81.6	74.8	64.5	52.1	42.6
		MCWB	37.1	37.1	44.7	51.1	60.1	67.2	69.3	69.6	65.5	57.4	50.9	40.8
	5%	DB	35.6	36.6	45.4	57.2	71.4	77.3	79.9	78.8	71.5	61.0	47.6	39.8
		MCWB	34.4	33.9	40.2	47.2	58.2	64.7	67.8	68.1	64.0	56.3	45.5	38.0
	10%	DB	33.5	33.6	40.9	53.0	67.5	74.1	77.3	75.9	68.3	57.6	43.8	37.1
		MCWB	32.5	32.0	36.5	43.7	56.2	63.0	66.4	66.7	62.7	52.9	41.3	35.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.4	44.1	51.4	59.7	66.0	73.0	74.2	75.3	69.5	62.6	54.5	45.9
		MCDB	46.1	44.7	54.6	65.7	74.6	81.7	81.8	83.3	73.9	65.5	55.4	46.2
	2%	WB	37.3	37.2	45.3	53.5	63.0	69.9	71.9	72.3	67.5	59.3	50.7	41.3
		MCDB	37.9	39.6	50.0	59.7	71.4	78.6	78.4	78.5	72.1	62.6	52.2	42.4
	5%	WB	34.7	34.4	41.0	48.3	60.7	67.4	70.4	70.4	66.0	56.9	46.3	38.1
		MCDB	35.5	36.2	44.6	54.4	67.8	73.9	76.6	75.7	70.2	60.5	47.2	39.5
	10%	WB	32.6	31.8	37.0	45.0	58.1	64.9	68.8	68.6	64.0	53.8	41.5	35.4
		MCDB	33.5	33.7	40.4	51.5	65.3	71.1	74.9	73.5	67.7	57.2	43.5	37.0
(35) Mean Daily Temperature Range	MDBR		21.5	23.8	21.0	19.7	22.6	22.3	21.3	21.4	20.6	17.3	13.9	17.6
	5% DB	MCDBR	17.6	16.2	22.1	26.9	29.7	26.1	24.5	23.6	24.2	20.7	16.0	14.2
		MCWBR	17.0	13.6	16.3	16.9	16.2	14.2	12.6	12.7	15.5	14.5	14.6	13.4
	5% WB	MCDBR	15.8	14.9	21.5	23.0	24.2	21.1	18.9	18.5	20.7	18.0	16.1	13.6
		MCWBR	15.7	13.1	17.1	16.7	14.8	13.0	11.0	11.4	14.6	14.1	15.6	13.5
(40) Clear-Sky Solar Irradiance	taub		0.279	0.289	0.315	0.344	0.365	0.396	0.401	0.396	0.366	0.346	0.312	0.277
	taud		2.378	2.319	2.351	2.378	2.405	2.334	2.357	2.363	2.436	2.461	2.498	2.464
	Ebn at Noon		264	283	291	290	286	276	273	270	270	259	250	254
	Edh at Noon		23	30	33	36	36	39	38	36	31	26	21	19
(44) All-Sky Solar Radiation	RadAvg		379	670	1130	1467	1746	1895	1922	1625	1246	735	397	275
	RadStd		49	66	107	128	165	141	99	92	121	86	57	26

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (22 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page